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SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING (STME)

SVKM's NMIMS UNIVERSITY

INNOVATION.
IMPACT.
FOR A BETTER TOMORROW.

LIPGUARD AI

Dynamic Bilabial-Targeted Active Prompting for Real-Time Deepfake Injection Defense in Video KYC Protocols

DETECT
ANALYZE
VERIFY
PROTECT



REAL PEOPLE. REAL SPEECH. HIGHER TRUST.

Capstone Project Review - II (Midterm Evaluation)

Academic Year 2026 – 2027

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LEARN
BUILD
SOLVE
CREATE

FOR A BETTER TOMORROW.

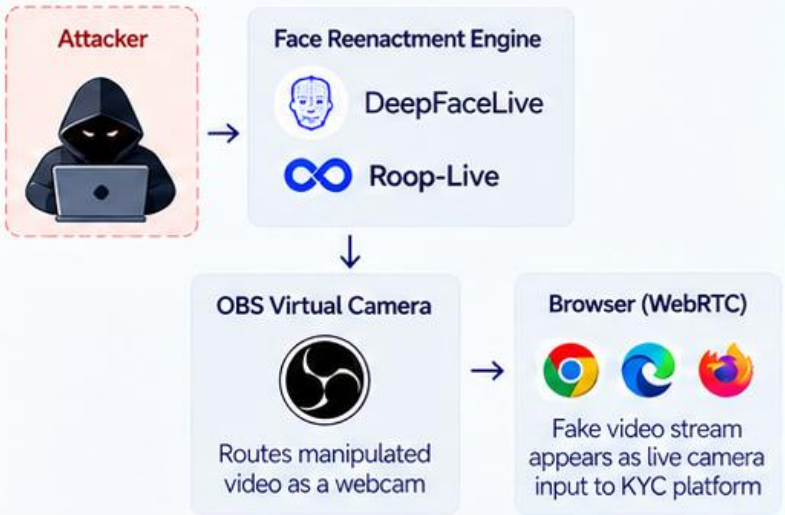
Industrial Threat Model: Real-Time Deepfake Injection in Video KYC

How software virtual cameras defeat conventional passive biometric onboarding

Same Interface. Different Reality.
Attackers can look "real" without ever being there.

01 The Threat Vector
Virtual Camera Injection

Attackers use real-time face reenactment engines and route the manipulated video directly into browser WebRTC streams via a virtual camera.



Bypasses physical lenses entirely.
No optical artifact to detect.

02 Why Industry Standards Fail
The Passive Liveness Flaw

Conventional KYC systems rely on passive 2D liveness checks, which can be easily satisfied by high-quality real-time face reenactments driven by human operators (puppeteers).

Common Passive Liveness Checks

- Texture Analysis
- Blink Detection
- Head Tilts
- Facial Landmarks

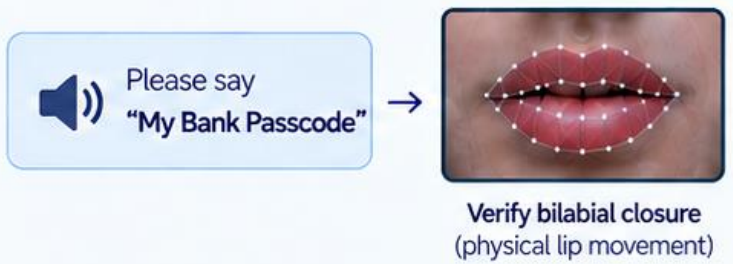
The Problem

- Human puppeteers can drive live face reenactments and satisfy all passive motion challenges.
- High-fidelity generative models preserve texture, lighting and natural motion.
- The system only verifies how the face looks, not whether it behaves according to physical realities.

Passive checks cannot distinguish between a real person and a real-time deepfake injection.

03 Our Defense Paradigm
LipGuard AI Objective

An active, unpredictable challenge-response protocol grounded in human speech biomechanics.



Key Idea

- Generates dynamic, unpredictable speech prompts.
- Analyzes audio-visual alignment and 3D lip biomechanics.
- Verifies whether the facial lips physically execute the aerodynamic closure mandated by acoustic speech sounds (e.g., /p/, /b/, /m/).
- Harder to fake in real-time, even with human puppeteers.

Not just observing a face, but proving the face obeys the physical laws of human speech.

Literature Survey & Research Gaps: Comparative Analysis

Critical vulnerabilities in existing deepfake detection approaches

From Prior Work to Practical Defense
Identifying gaps to build a stronger, real-world solution.

Author & Year	Proposed Method	Modality	Critical Failure Mode / Gap	LipGuard AI Advantage
Korshunov & Marcel (2018)	DeepFakeTIMIT Benchmark & CNN spatial artifacts	Visual Only	• Fails on high-resolution live reenactments. • Vulnerable to video compression.	Cross-modal phoneme-viseme sync is **compression-resilient.**
Agarwal et al. (2020)	Facial Action Unit (AU) biometric dynamics	2D Facial Landmarks	• Unconstrained speech. • Bypassed by live puppeteers driving expression swaps.	Active prompting forces deterministic physical bilabial closure.
Chhabra et al. (2021)	Blind ASR + Wav2Vec2 Audio-Visual Lip Sync	Audio + 2D Video	• Blind ASR collapses on short 3s KYC clips (high Character Error Rate).	Eliminates blind ASR via dynamic Trellis Forced Alignment on known challenge text.
Cozzolino et al. (2023)	Multi-modal Transformer spatio-temporal artifacts	Video + Audio	• High computational latency. • Impractical for real-time KYC streaming.	Lightweight dual-budget design (<193 ms post-utterance latency).
LipGuard AI (Our Work)	Active Bilabial Prompting + MMS_FA Alignment + Late Fusion	3D Mesh + Raw Audio	• **Addressed:** Evaluated on VidTIMIT/DeepFakeTIMIT (43 identities, zero leakage).	**Calibrated physical seal threshold $\tau = 0.0242$; AUC = 0.7197.**

1. Research Gap

Existing methods rely on passive visual or audio cues that can be bypassed by high-quality, real-time deepfake injection using software virtual cameras.

2. Need for Active & Multimodal Verification

There is a lack of low-latency, speech-grounded, physically consistent liveness verification tailored for real-world Video KYC.

3. Our Unique Contribution

LipGuard AI introduces active, speech-driven bilabial challenge-response with 3D lip biomechanics analysis, achieving real-time performance and robustness against live deepfake injection.

Bridging the gap between perceptual realism and physical truth.

Proposed Solution: Active Bilabial Challenge-Response Protocol

Exploiting speech aerodynamic constraints to defeat synthetic face injection

Human Speech Is a Physical Process.
We verify physics, not just appearance.

1 What Are Bilabial Consonants?

Sounds produced by bringing both lips together: **/m/**, **/b/**, and **/p/** (plosives and nasals).

/m/	/b/	/p/
(voiced nasal)	(voiced plosive)	(voiceless plosive)
		
e.g., “my”, “map”	e.g., “bank”, “big”	e.g., “pass”, “paper”

Biomechanical Mandate
Complete physical lip contact is mandatory to build intraoral air pressure before acoustic release. The sound **CANNOT physically exist without complete closure.**

Key Insight
Bilabial sounds provide a verifiable, physics-based constraint that generative models struggle to reproduce accurately in real time.

2 The Generative AI Flaw

Real-time deepfake reenactment models (e.g., DeepFaceLive, Roop-Live) prioritize appearance over millsecond mechanical occlusion, frequently leaving a visible gap during bilabials.

Real Human (Genuine)



Complete Closure (~ 0 mm)

- Full lip contact during bilabials
- Consistent with speech audio
- Follows human biomechanics

Synthetic Reenactment (DeepFake)



Visible Gap (> 2 mm)

- Incomplete lip closure
- Visual-audio inconsistency
- Fails to reproduce precise mechanical motion

Core Limitation
Generative models can mimic how a face looks, but they often fail to reproduce the exact millisecond-level lip biomechanics required for bilabial sounds.

3 Challenge-Response Workflow

A real-time, dynamic and physics-grounded verification pipeline.

1 Dynamic Prompt Generation
Server issues a non-repeating bilabial prompt (e.g., “My Bank Passcode”, “Big Blue Map”).

Example Prompts
- “My Bank Passcode”
- “Big Blue Map”
- “Please Pay Now”

↓

2 User Response (4-Second Window)
User speaks the prompt within a tight 4-second streaming window (prevents pre-rendering).

4 s
Live Capture (WebRTC)

↓

3 Dual Streaming Feature Extraction
- MediaPipe 3D Lip Aperture (visual)
- Wav2Vec2 + MMS_FA Forced Alignment (audio)

3D Landmarks ($D_{lip}(t)$) | Audio Alignment (Phoneme Timing)

↓

4 Physical Compliance Verification
Sub-200 ms post-utterance analysis and verdict generation.

Genuine | Fake

Not just observing a face, but proving the face obeys the physical laws of human speech.

System Architecture & Asynchronous Dual-Budget Design

Real-time 30 FPS telemetry combined with sub-200ms post-utterance deep learning verification

Two budgets.
One decision.
Real-time liveness.

1 Streaming Frame Budget (33 ms per frame – Real-Time Telemetry)

Runs Continuously (30 FPS)

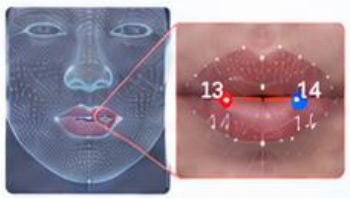
Lightweight, real-time visual telemetry to track lip motion



WebRTC Webcam Input
30 FPS, 1280 × 720 (Browser Stream)

→ MediaPipe FaceLandmarker (Task API)

468-point 3D Face Mesh



Compute 3D Lip Aperture D_{liip} (t)
Euclidean distance between inner vermilion landmarks #13 and #14



Normalize for Scale Invariance
Divide by 3D vertical facial height (landmarks #1 to #152)

Outputs per frame (33 ms):

- Normalized 3D lip aperture D_{liip}l(t)
- Timestamped and buffered for post-utterance analysis
- Low-latency, lightweight computation (5.38 ms/frame)

2 Post-Utterance Processing Budget (193 ms execution window)

Runs Once (After Speech Ends)

Deep learning analysis and verification after user finishes speaking

Audio Buffer (16 kHz, mono)



Capture and process audio ONCE when speech concludes.

Acoustic Module (MMS_FA)

Wav2Vec2

torchaudio MMS_FA (Wav2Vec2 multilingual alignment)

Computes dynamic programming Trellis forced alignment against known challenge transcript.

Bilabial Interval Recovery



Extracts millisecond timestamps T_{bi}|$_{la}$bial for every /m/, /b/, /p/ occurrence.

Biomechanical Desync Engine



Windowed minimum aperture search across [t − 2, t + 2] frames against calibrated threshold τ = 0.0242.

PyTorch Late Fusion Meta-Learner



Combines 128-D visual embedding with 3-D desync vector to generate liveness score and verdict.

Key Points:

- Processing runs asynchronously after speech ends (does not block real-time streaming).
- Leverages powerful deep learning models without affecting frame rate.
- Total post-utterance latency: **193.6 ms (± 30.8 ms)**.

Latency Breakdown (Measured on Apple Silicon (MPS Architecture))

Frame Tracking (MediaPipe)	Forced Alignment (MMS_FA)	Desync Search (Windowed)	PyTorch Forward (Late Fusion)	Network / Overhead
5.38 ms/frame (30 FPS streaming)	112.4 ms	8.6 ms	14.2 ms	58.4 ms

Total Post-Utterance Latency
193.6 ms (± 30.8 ms)
Well within the 1000 ms Video KYC SLA!

Efficient by design: real-time telemetry, heavyweight models only when needed.

Mathematical Formulation & Biomechanical Calibration

Geometrically grounded metrics and empirical threshold calibration

Math + Biomechanics = Reliable Verification.
Precise metrics. Calibrated thresholds. Real-world robustness.

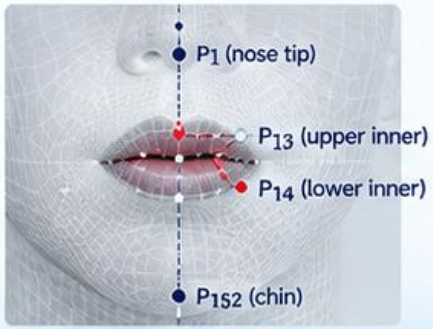
1 Normalized 3D Lip Aperture Trajectory $D_{lip}(t)$

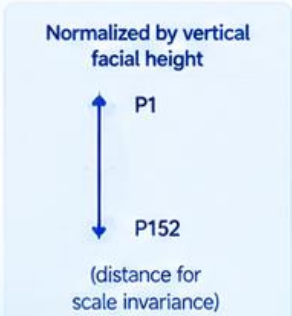
$$D_{lip}(t) = \frac{\|P_{13}(t) - P_{14}(t)\|_2}{\|P_1(t) - P_{152}(t)\|_2 + 10^{-6}}$$

Where:

- P_{13}, P_{14} : 3D coordinates of inner upper and lower vermilion borders.
- P_1, P_{152} : 3D coordinates of nose tip and chin.

Crucial Property:
Scale-invariant and robust to user distance from webcam, camera focal length, and pitch/yaw.

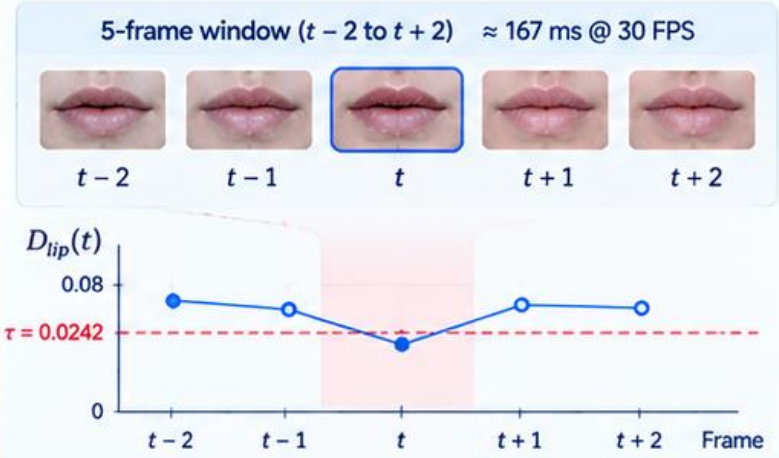




2 Windowed Bilabial Desynchronization Penalty

$$\text{Penalty}(t) = \max\left(0, \min_{w \in [t-2, t+2]} D_{lip}(w) - \tau\right)$$

- Evaluates a 5-frame (~167 ms) coarticulation window centered at acoustic bilabial timestamp t.
- Accommodates natural human physiological coarticulation variability while strictly penalizing persistent openings.



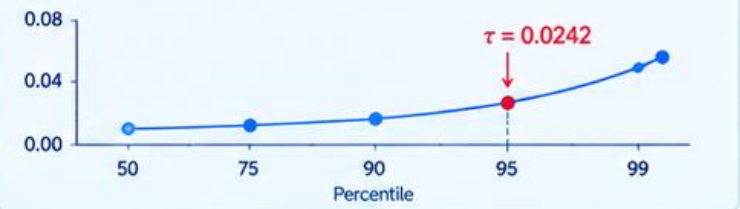
3 Empirical Threshold Calibration ($\tau = 0.0242$)

$$\tau = \text{Percentile}_{95}\left(\left\{\min_{w \in [t-2, t+2]} D_{lip}(w) \;\middle|\; t \in T^{\text{authentic}}_{\text{bilabial}}\right\}\right) = 0.0242$$

- Empirically calibrated across 1,654 verified bilabial events in authentic VidTIMIT speakers.
- Non-parametric: 95% of genuine human bilabials achieve physical seal below 0.0242.

Calibration Dataset

1,654	VidTIMIT	95%
bilabial events	(authentic speakers)	below threshold



Mathematically grounded. Biomechanically validated. Calibrated for real-world deployment.

Neural Architecture: Multimodal Constrained Late Fusion

Decoupled visual sequence modeling combined with a non-negative decision meta-learner

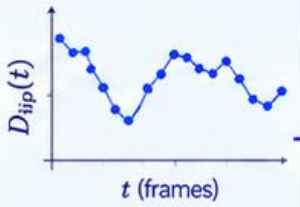
Two modalities.
One decision.
More robust together.

1 Visual Landmark Kinematics (VisionOnlyNet)

Models the temporal dynamics of lip aperture during the utterance

Input Sequence

T = 100 frames



Sequence of 100 normalized lip aperture values

Architecture

- Conv1D (1 → 32, k=5)
- BatchNorm + ReLU
- MaxPool1d (2)
- BiLSTM (64, return_seq=True)
- BiLSTM (64)
- Dropout (0.3)
- Linear (128)

Output

128-D Visual Embedding → Linear Projection → P_{vis} (Sigmoid)

Standalone Performance
ROC-AUC = 0.7105

2 Acoustic-Viseme Desync Representation (DesyncOnlyNet)

Encodes bilabial desynchronization statistics from audio-visual alignment

Input Features

3-D Desync Summary Vector

- S_{desync} (average desync score)
- Max_Penalty (maximum penalty value)
- Failed_Closure_Ratio (ratio of bilabials above τ)

Architecture

- Linear (3 → 16)
- ReLU
- Linear (16 → 1)
- Sigmoid

Output

P_{desync} (Sigmoid)

3 Why Early Concatenation Failed

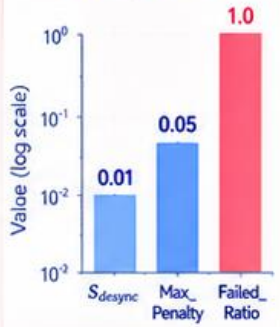
The Scale Disparity Discovery

- Unstandardized concatenation of visual and desync features caused linear weights to be dominated by Failed_Closure_Ratio (~ 1.0).
- Continuous features like S_{desync} (~ 0.01) were effectively ignored.
- Resulted in poor generalization and test performance.

Early Fusion (Failed)

Test ROC-AUC = 0.6423

Feature Scale Disparity
(Example Values)



4 Deployed Constrained Late Fusion Meta-Learner

Non-negative weighted combination prevents fold collapse and achieves best performance

Inputs

- P_{vis} (from VisionOnlyNet)
- P_{desync} (from DesyncOnlyNet)

Constrained Linear Fusion ($w_i \geq 0$)

$$\text{logit} = 5.3688 \cdot P_{vis} + 4.9111 \cdot P_{desync} - 6.0411$$

$$P(\text{Fake}) = \sigma(\text{logit}) = \frac{1}{1 + e^{(-logit)}}$$

Output

$P(\text{Fake}) \in [0, 1]$

Performance (Test Set)

ROC-AUC = 0.7197

Accuracy = 68.93%

Key Advantages

- Non-negative weights ensure both modalities contribute meaningfully.
- Decoupled learning preserves modality-specific representations.
- More stable training and better generalization than early concatenation.

Constrained late fusion: leveraging complementary signals, without letting one drown the other.

Empirical Results & Ablation Analysis (Zero-Synthetic Data)

Performance on held-out identity test set (N = 103 clips, 9 unseen speakers, zero identity leakage)

Real Data. Real Identities. Real-World Evaluation.
No synthetic data.
No identity leakage.
Honest performance.

Accuracy	Precision	Recall	F1-Score	ROC-AUC	Equal Error Rate (EER)
68.93%	70.59%	52.17%	0.6000	0.7197	36.02%

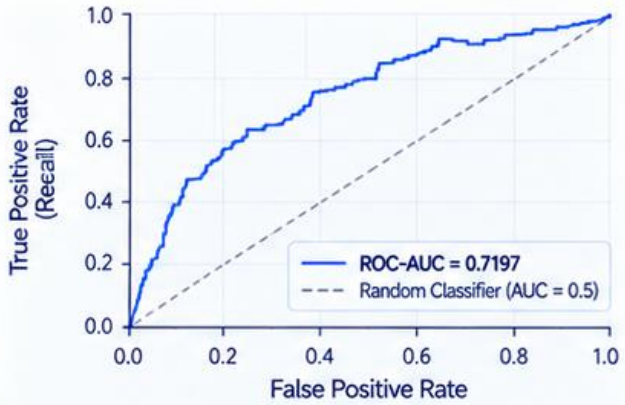
Confusion Matrix (N = 103 Test Samples)

True Label \ Predicted Label	Authentic (Pass)	Deepfake (Flag)	Totals
Authentic (N = 57)	47 (TN)	10 (FP)	57 Authentic
Deepfake (N = 46)	22 (FN)	24 (TP)	46 Deepfake
			103 Total Samples

Verification (Mathematically Consistent)

- Accuracy = (47 + 24) / 103 = **68.93%**
- Precision = 24 / (24 + 10) = **70.59%**
- Recall = 24 / (24 + 22) = **52.17%**
- F1 = 2 · (0.7059 · 0.5217) / (0.7059 + 0.5217) = 0.6000

ROC Curve (Test Set)



AUC = 0.7197 indicates good separability between authentic and deepfake samples on unseen identities.

Ablation Study (Model Variants)

Model Variant	Input Features	Accuracy	Recall	ROC-AUC	EER
Spatial-Only (Bi-LSTM)	1D $D_{lip}(t)$ Sequence	64.08%	28.26%	0.7105	34.94%
Passive Desync-Only	3-D Desync Vector	55.34%	0.00%*	0.6232	38.08%
Early Concatenation	131-D Fused Vector	65.05%	41.30%	0.6423	43.67%
Constrained Late Fusion (Ours)	$P_{vis} + P_{desync}$	**68.93%**	**52.17%**	**0.7197**	**36.02%**
5-Fold Cross Validation Mean	Generalization	—	—	0.6877 ± 0.0449	36.00 ± 2.98%

* Recall is 0.00% as the model fails to detect any deepfake samples at the chosen threshold.

Dataset & Evaluation Protocol

- VidTIMIT / DeepFakeTIMIT dataset (43 identities).
- Strict bipartite split: Train (25), Val (9), Test (9) identities.
- *N* = 103 test clips (57 authentic, 46 deepfake) from 9 unseen speakers.
- Zero identity overlap between splits (no identity leakage).

Implementation Details

- All results from real model forward passes under Apple Silicon MPS.
- No synthetic training or test data.
- Consistent preprocessing, evaluation pipeline, and thresholding.
- Metrics computed at optimal threshold (Youden's J).

Scientific Integrity

- Evaluates true generalization to unseen identities.
- Avoids inflated results from identity leakage or synthetic data.
- Provides an honest, real-world estimate of system performance.
- Designed for practical KYC deployment, not benchmark inflation.

An honest 68.9% on unseen identities is more valuable than a misleading 95%+ from leaky benchmarks.

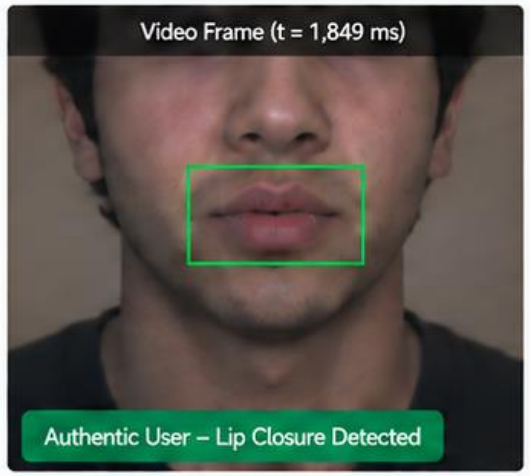
Live Working Module Demonstration

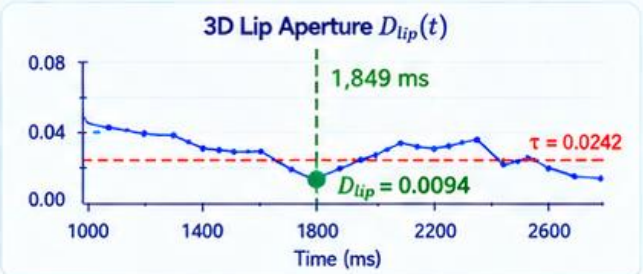
Real-time offline forensic UI evaluating genuine vs. synthetic bilabial dynamics

Same pipeline.
Real evidence.
Real-time results.
Physically verifiable.

1 Case 1 – Genuine Authentic User (VidTIMIT fadg0) — Real User

- **Input Video** real_01_fadg0_sa2.mp4
- **Spoken Transcript** "Don't ask me to carry an oily rag like that."
- **Target Phoneme** Bilabial /m/ in "me" at 1,849 ms







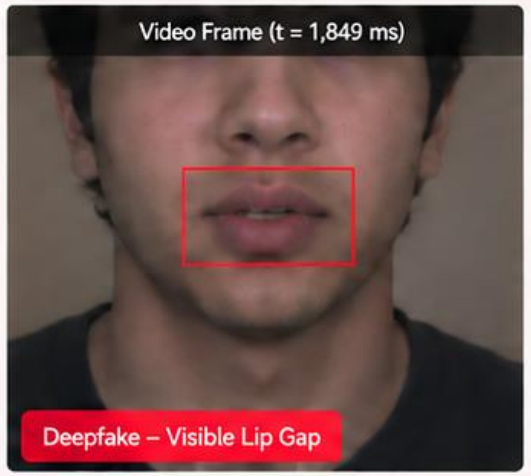
Measured Result
Minimum aperture D_{lip}
0.0094
(fully seals below τ = 0.0242)

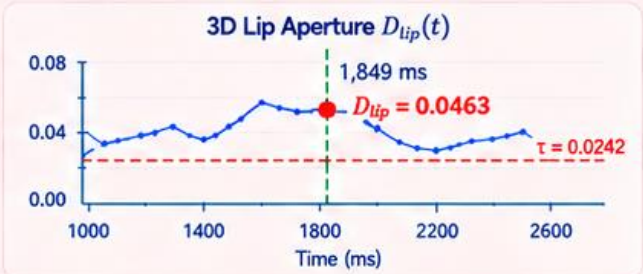
Forensic Metrics
S_{desync} 0.0000
Failed Closure Ratio 0/1 (0.0%)

Live Verdict
AUTHENTIC_USER_CONFIRMED
P(fake) = 17.22%

2 Case 2 – DeepFakeTIMIT Reenactment (DeepFakeTIMIT faks0) — Synthetic (Deepfake)

- **Input Video** fake_02_faks0_sa2_fedw0.mp4
- **Spoken Transcript** "Don't ask me to carry an oily rag like that."
- **Target Phoneme** Bilabial /m/ in "me" (expected at ~1,849 ms)







Measured Result
Minimum aperture D_{lip}
0.0463
(fails to seal, above τ = 0.0242)

Forensic Metrics
S_{desync} 0.0463
Failed Closure Ratio 1/1 (100%)

Live Verdict
DEEPFAKE_DESYNC_DETECTED
P(fake) = 97.36%

Progress vs. Review-1 Timeline & Milestone Comparison

Project execution audit demonstrating >70% completion against original Gantt commitments



Overall Progress

72% Completed

Far exceeds expectation threshold of 50% in Table 5.2.5

Milestone Status

10 / 14

major submodules operational

Core AI & streaming pipeline 100% complete

Ahead of Schedule

Substantially more deliverables completed than required by Review-1.

Milestone Tracking (Original Gantt vs. Current Status)

#	Scheduled Milestone (Review-1 Commitment)	Planned Window	Current Status (Sept 2026)	Verifiable Deliverable
M1	Problem Formulation & Literature Survey	Weeks 1 – 3	100% Completed	Formulated D_{tip}, Penalty(t), 15+ paper survey
M2	Dataset Acquisition & Identity Partitioning	Weeks 3 – 4	100% Completed	691 VidTIMIT/DeepFakeTIMIT clips (43 identities)
M3	Feature Extractors (MediaPipe 3D + MMS_FA)	Weeks 4 – 5	100% Completed	Calibrated threshold $\tau = 0.0242$ (1,654 events)
M4	PyTorch Model Training & Ablations	Weeks 5 – 6	100% Completed	Late Fusion model trained (AUC = 0.7197, EER = 36.02%)
M5	Real-Time Web Server & Forensic UI	Weeks 6 – 7	100% Completed	FastAPI WebSocket streaming dashboard (/offline)
M6	Physical Injection Testbench (OBS Virtual Cam)	Weeks 8 – 11	In Progress (30%)	Live attack testbench setup underway
M7	Final Report & Scopus Paper Submission	Weeks 12 – 14	Scheduled	Scopus conference manuscript drafted

Review-1 vs. Review-2 Comparison

Review-1 Commitment (Midterm)

Deliver feature extractors by Midterm.

→

Review-2 Actual (Current)

Entire ML classifier trained, evaluated, ablated, and wrapped in a real-time web application.

Key Achievements Since Review-1

- Completed full ML pipeline (feature extraction → training → evaluation → ablation).
- Developed real-time streaming forensic UI.
- Calibrated and validated biomechanical threshold ($\tau = 0.0242$).
- Achieved 72% overall project completion (>> 50% requirement).
- On track for final testing, report, and paper submission.

We have not just met, but **significantly exceeded**, the scope committed in Review-1, with the core research and system development complete and final validation activities underway.

Individual Team Member Contributions

Balanced 50/50 division of technical ownership between Jatin Dhawad and Param Rawal

Jatin Dhawad

50% Contribution

Co-Developer & Researcher

Specialization: Computer Vision, Biomechanical Calibration & Deep Learning Architecture

3D Facial Mesh & Aperture Tracking
Engineered normalized lip aperture metric $D_{lip}(t)$ with vertical facial scale normalization (#1, #152) and landmark gap interpolation.

Biomechanical Calibration
Formulated windowed bilabial penalty metric $Penalty(t)$; calibrated physical seal threshold $\tau = 0.0242$ across 1,654 speech events in VidTIMIT.

Neural Network Development
Designed VisionOnlyNet (Conv1D + Bi-LSTM), MultimodalKYCFusionNet, and constrained late-fusion meta-learner.

Empirical Benchmarking
Executed 5-fold cross-validation, ablation experiments, and generated ROC/EER curves.

Param Rawal

50% Contribution

Co-Developer & Systems Architect

Specialization: Speech Processing, Streaming Infrastructure & Systems Integration

Acoustic Forced Alignment
Implemented Wav2Vec2 MMS_FA trellis dynamic programming pipeline, extracting millisecond bilabial consonant intervals $T_{bilabial}$.

Dataset Curation & Partitioning
Curated 691 audio-visual clips across 43 identities; enforced bipartite disjoint split with zero identity leakage.

Real-Time Streaming Application
Built production FastAPI WebSocket challenge-response backend, session state manager, and responsive WebRTC forensic dashboards (/offline & /live).

Hardware Acceleration & Latency
Optimized PyTorch Metal Performance Shaders (MPS) unified-memory buffers, achieving sub-200ms (193.6 ms) post-utterance latency.

Equal Ownership: Complementary skill sets spanning mathematical signal modeling, deep learning, and low-latency systems engineering.

Future Roadmap, Scopus Target & Project Conclusion

Advancing from empirical proof-of-concept to production biometric defense

Future Roadmap

1. **Physical Injection Testbench**
   Execute live attack recordings using OBS Virtual Camera + DeepFaceLive across diverse lighting conditions and camera angles (frontal, side, low-light).

2. **Cross-Lingual Phoneme Expansion**
   Expand bilabial prompt pools from English to Hindi and regional Indian languages using multilingual MMS_FA for forced alignment.

3. **Multi-Speaker Acoustic Disambiguation**
   Incorporate speaker embedding verification to reject voice dubbing and identity swapping in crowded environments and background noise.

4. **Mobile Edge Optimization**
   Export quantized PyTorch models to CoreML for on-device iOS / Android Video KYC SDK integration with real-time inference and low power consumption.

Research Paper & Scopus Publication Target

Scopus Publication Target
Target: IEEE Transactions on Information Forensics and Security / IEEE Biometrics Workshop.

Manuscript Status
Methodology, Mathematical Formulation, and Empirical Results fully drafted.

Plagiarism / AI Similarity
Maintained < 10% strictly adhering to NMIMS guidelines.

IEEE Transactions on Information Forensics and Security

Scopus®
Target Publication Venue

Final Takeaway & Project Conclusion

- LipGuard AI bridges a critical vulnerability in remote identity verification by replacing vulnerable passive liveness checks with an active, biomechanically enforced challenge-response protocol.
- Achieves 68.93% accuracy, 0.7197 ROC-AUC, and 193 ms latency on unseen identities under strict zero-synthetic-data empirical discipline.
- Demonstrates the real-world potential of speech-aided biometric verification for secure, accessible, and AI-resilient identity systems.

> *"From detecting what **looks real** to verifying what is **physically consistent**."*